

Practical 9 DAA

Summary

Prim's Algorithm is a **greedy algorithm** used to find the **Minimum Spanning Tree (MST)** of a connected, weighted, undirected graph. It starts with one vertex and repeatedly selects the minimum-weight edge that connects a selected vertex to an unselected vertex. The Python implementation uses an **adjacency matrix** and a `selected` array to keep track of vertices already included in the MST.

Conclusion

Prim's Algorithm provides an efficient way to construct a Minimum Spanning Tree while ensuring that the total edge weight is minimized. The given Python program successfully finds the MST and calculates its total weight. The adjacency-matrix implementation has a **time complexity of $O(V^2)$** , making it suitable for graphs with a moderate number of vertices.